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PHY201 Exam 5 on chapter(s): 11, 12

Name: \_\_\_\_\_

These questions assess your understanding of the material from Chapter(s) 11, 12 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Ashley attaches a nozzle of diameter 6.9 mm to a garden hose of diameter 2.3 cm. If the hose fills up a 2-L bucket in 12.8 seconds, what is the speed of the water that leaves the nozzle?

Nozzle: 2  
 6.9 mm  
 $V_2 = ?$   
 Hose: 1  
 $d = 2.3 \text{ cm}$   
 2L bucket  
 12.8 s  
 $Q_1 = A_1 V_1$   
 $Q_2 = A_2 V_2$   
 $Q_1 = Q_2$   
 $A_1 V_1 = A_2 V_2$   
 $V_2 = \frac{A_1 V_1}{A_2}$   
 $V_2 = \frac{\pi (0.0115)^2 (0.0159 \text{ m})}{\pi (0.00345)^2}$   
 $V_2 = 4.18 \text{ m/s}$

ANSWER: 4.18 m/s ✓

2. Grace attaches a nozzle of diameter 6.8 mm to a garden hose of diameter 3.0 cm. If the hose fills up a 4-L bucket in 11.6 seconds, calculate the pressure in the hose assuming level, frictionless flow.

Nozzle: 2  
 $d = 6.8 \text{ mm}$   
 $P_2 = 101325 \text{ Pa}$   
 $V_2 = ?$   
 Hose: 1  
 $d = 3 \text{ cm}$   
 4L in  
 11.6 s  
 $P_1 = ?$   
 $Q_1 = A_1 V_1$   
 $Q_2 = A_2 V_2$   
 $Q_1 = Q_2$   
 $A_1 V_1 = A_2 V_2$   
 $V_1 = \frac{A_2 V_2}{A_1}$   
 $V_1 = \frac{\pi (0.0034)^2 (9.498 \text{ m/s})}{\pi (0.015)^2}$   
 $V_1 = 0.48779 \text{ m/s}$   
 $P_1 + \frac{1}{2} \rho V_1^2 = P_2 + \frac{1}{2} \rho V_2^2$   
 $P_1 + \frac{1}{2} (1000) (0.48779)^2 = 101325 + \frac{1}{2} (1000) (9.498)^2$   
 $P_1 + 118.97 = 146436.701$   
 $P_1 = 146317.7 \text{ Pa}$

ANSWER: 146317.7 Pa ✓

3. Kevin's doctor performs a special test and tells him that a 2.9-N force can rupture his eardrum. If his eardrum has an area of  $1.0 \text{ cm}^2$ , at what depth in freshwater would his eardrum rupture, assuming the gauge pressure in the middle ear is zero and he does nothing to alleviate the pressure?

$F = 2.9 \text{ N}$   
 $A = 1 \text{ cm}^2$   
 $h = ?$   
 $P = \frac{F}{A}$   
 $P = \rho g h$   
 $h = \frac{P}{\rho g}$   
 $h = \frac{29000 \text{ Pa}}{1000 \text{ kg/m}^3 \cdot 9.8 \text{ m/s}^2}$   
 $h = 2.96 \text{ m}$

ANSWER: 2.96 m ✓

+3

4. The aorta pumps blood from the heart to the rest of the body. Calculate the average speed of the blood in Grace's aorta if the flow rate of her blood is 11.5 L/min. Grace's aorta has a radius of 10 mm.

$$Q = 11.5 \text{ L/min}$$

$$r = 10 \text{ mm}$$

$$1 \text{ L} = 1 \times 10^{-3} \text{ m}^3$$

$$= \frac{0.115 \text{ m}^3}{60} = 1.92 \times 10^{-4} \text{ m}^3/\text{s}$$

$$A = \pi r^2$$

$$= \pi (0.01)^2$$

$$= 3.14 \times 10^{-4}$$

$$Q = AV$$

$$1.92 \times 10^{-4} = 3.14 \times 10^{-4} (V)$$

$$V = 0.611 \text{ m/s}$$

$$\text{ANSWER: } 0.611 \text{ m/s}$$



5. A particular dam is 590-m wide and contains freshwater that is 103-m deep. What is the average pressure on the dam due to the water?

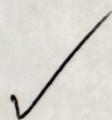
$$h = 103 \text{ m}$$

$$P = \rho gh$$

$$P = \frac{1000(9.8)(103)}{2}$$

$$P_{\text{avg}} = 504700 \text{ Pa}$$

$$\text{ANSWER: } 504700 \text{ Pa}$$



6. The aorta pumps blood from the heart to the rest of the body. The flow rate of blood through Jonathan's aorta is 13.2 L/min and Jonathan's aorta has a radius of 10 mm. Recall that blood flows through smaller blood vessels known as capillaries, which have an average diameter of 8.1  $\mu\text{m}$ . The speed of blood in the capillaries is about 0.29 mm/s. Calculate the number of capillaries in Jonathan's blood circulatory system.

$$Q = 13.2 \text{ L/min}$$

$$Q_{\text{aorta}} = 1$$

$$r = 10 \text{ mm}$$

$$Q_{\text{capillaries}} = 2$$

$$d = 8.1 \mu\text{m}$$

$$V = 0.29 \text{ mm/s}$$

$$n_{\text{cap}} = ?$$

$$1 \text{ L} = 1 \times 10^{-3} \text{ m}^3$$

$$\frac{13.2 \times 10^{-3}}{60} = 2.2 \times 10^{-4} \text{ m}^3/\text{s}$$

$$A_1: \text{aorta}$$

$$A_2: \text{cap}$$

$$A = \pi (0.01)^2$$

$$= 3.14 \times 10^{-4}$$

$$r = 4.05 \times 10^{-6}$$

$$A = \pi (4.05 \times 10^{-6})^2$$

$$= 5.15 \times 10^{-11}$$

$$Q = AV_1$$

$$2.2 \times 10^{-4} = 3.14 \times 10^{-4} (V_1)$$

$$V_1 = 0.7 \text{ m/s}$$

$$V_2 = 0.29 \text{ mm/s}$$

$$= 2.9 \times 10^{-4} \text{ m/s}$$

$$\text{ANSWER: } 1.47 \times 10^{10} \text{ capillaries}$$

$$A_1 V_1 = A_2 V_2$$

$$3.14 \times 10^{-4} (0.7) = n_2 (5.15 \times 10^{-11}) (2.9 \times 10^{-4})$$

$$2.198 \times 10^{-4} = n_2 (1.4935 \times 10^{-14})$$

$$n_2 = 1.47 \times 10^{10}$$

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7. Calculate the depth below the surface of freshwater at which the pressure due to the weight of the water equals 1.37 atm.

ANSWER: 14.16m

$$P = 1.37 \text{ atm}$$

$$1 \text{ atm} = 101325 \cdot 1.37 = 138815.25 \text{ Pa}$$

$$P = \rho g h$$

$$138815.25 = 1000(9.8)(h)$$

$$h = 14.16 \text{ m}$$



8. A freshwater reservoir has a surface area of 42 km<sup>2</sup> and an average depth of 59 m. What mass of water is held behind the dam?

ANSWER:  $2.478 \times 10^{12} \text{ kg}$

$$A = 42 \text{ km}^2$$

$$\frac{42 \text{ km}^2}{1 \text{ km}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot 59 = 2478000000 \text{ m}^3$$

$$\rho = \frac{m}{V}$$

$$1000 = \frac{m}{2478000000}$$

$$m = 2.478 \times 10^{12}$$



9. Calculate the maximum buoyant force that saltwater could exert on a ship with a mass of  $1.1 \times 10^8 \text{ kg}$  if the ship displaces  $3.9 \times 10^6 \text{ m}^3$  of water. Saltwater has a density of  $1025 \text{ kg/m}^3$ .

ANSWER:  $3.92 \times 10^{10} \text{ N}$

$$P = \rho g h \quad P = \frac{F}{A}$$

$$AP = F$$

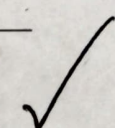
$$F_b = \rho V g$$

$$F_b = 1025(3.9 \times 10^6)(9.8)$$

$$= 3.92 \times 10^{10} \text{ N}$$

$$V = 3.9 \times 10^6 \text{ m}^3$$

$$\rho = 1025 \text{ kg/m}^3$$



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1. 4.18 m/s
2. 146.28 kPa
3. 2.96 m
4. 61.01 cm/s
5. 504.7 kPa
6.  $1.47 \times 10^{10}$  capillaries
7. 14.16 m
8.  $2.48 \times 10^{12}$  kg
9.  $3.92 \times 10^{10}$  N